

Tools of the Trade: Linux and SQL

Module 1

Introduction to operating systems

Module 2

The Linux operating system

All about Linux

Welcome to module 2

Video • 15 min

Introduction to Linux

Video • 3 min

What Linux and give in the cybersecurity field

Video • 2 min

Linux architecture

Video • 3 min

Linux architecture explained

Reading • 8 min

Engine Components of the Linux OS

Upgraded App Item • 10 min

Test your knowledge: All about Linux

Practice Assignment • Grade: 100%

Linux distributions

Linux Distributions

Video • 2 min

Kali Linux™

Video • 2 min

More Linux distributions

Reading • 4 min

Package managers for installing applications

Reading • 8 min

Resources for completing Linux labs

Reading • 4 min

Lab tips and troubleshooting steps

Reading • 4 min

Activate Intel® software in a Linux distribution

Upgraded App Item • 30 min

Optional Example: Install software in a Linux distribution

Upgraded App Item • 10 min

Example: Install software in a Linux distribution

Reading • 3 min

Test your knowledge: Linux distributions

Practice Assignment • 4 min

The shell

Introduction to the shell

Video • 4 min

Different types of shells

Reading • 4 min

Input and output in the shell

Video • 2 min

Activate Bash® interactive input and output in the shell

Upgraded App Item • 30 min

Optional Example: Terminate input and output in the shell

Upgraded App Item • 10 min

Example: Bash® interactive input and output in the shell

Reading • 10 min

Test your knowledge: The shell

Practice Assignment • 4 min

Review: The Linux operating system

Wrap-up

Video • 10 sec

Glossary/terms from module 2

Reading • 2 min

Module 2 challenge

Graded Assignment • 40 min

Module 3

Linux commands in the Bash shell

Module 4

Databases and SQL

Resources for completing Linux labs

0:01 / 0:22

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Qwiklabs has updated their terms of services to include an **age requirement of 18+ to use the platform**, in order to comply with regulations in the US and EU. Learners without access to Qwiklabs are still able to complete the certification and gain the badge by meeting the Qwiklabs instructions, examples, and participating in other hands-on activities throughout the certificate. This participation is oriented to understanding the certificate's concepts and preparing learners for graded assessments.

This course features hands-on lab activities where you'll have the opportunity to practice Linux commands in the terminal. You'll use a platform called Qwiklabs to complete these labs. In this reading, you'll learn how to use Qwiklabs.

This reading first provides a section on how to use Qwiklabs, which includes details on how to launch a lab, how to interact within the Qwiklabs environment, and how to end a lab. This is followed by another section on helpful navigation tips and keyboard shortcuts; these may be useful when working in the terminal.

Note: You will not launch Qwiklabs directly from this reading and instead will do this through lab activities and exemplars that you encounter throughout the course.

How to use Qwiklabs

Launching Qwiklabs

When you select a lab, you start from a Coursera page. You will need to click **Launch App** on that page. After you click **Launch App**, a new tab will open with a Qwiklabs page that contains instructions for that particular lab.

Start Lab button

On the Qwiklabs page, you must click **Start Lab** to open a temporary terminal. The instructions for the lab will move to the right side of the screen.

Start Lab

00:29:00

Read the instructions and complete all the tasks in the lab by entering commands in the terminal.

Note: It may take a moment for the terminal to start.

Lab control dialog box

After you click **Start Lab**, the lab control dialog box opens. It contains the **End Lab** button, the **timer**, and the **Open Linux Console** button.

You can hide or unhide the dialog box by clicking the following icon in the red box:

End Lab

00:58:16

Optional: After you click in the terminal, do not close the window. Instead, click the **Open Linux Console** button to open a new browser window.

Open Linux Console

The timer

The **timer** starts when the terminal has loaded. The timer keeps track of the amount of time you have left to complete a lab. The timer count down until it reaches 00:00:00. When it does, your temporary terminal and resources are deleted.

You will have ample time to complete the labs. But, stay focused on completing the tasks to ensure you enjoy your time well.

Open Linux Console button

When you click the button to **Open Linux Console**, the terminal opens in a new browser window:

End Lab

00:58:16

Use this feature if you want a full-screen view of the terminal. You can close this window at any time. Closing the window does not end your lab, and you can continue working in the terminal in the original tab.

Check progress

You can check your progress by clicking **Check my progress** at the end of each task.

Use the echo command

Check my progress

If you haven't yet completed a task, you'll receive hints on what you must do to complete it.

You can click **Check my progress** whenever you want to check the completion status of a task or receive a hint.

Using copy/paste commands

The first time you try to use copy or paste keyboard shortcuts (such as **CTRL + C**), you'll receive a pop-up requesting permission to use your device's clipboard. **"gnomecoursewarewikilabs.com wants to see text and images copied to the clipboard."** Please click **Allow** if you would like to be able to use these shortcuts in the Qwiklabs platform. If you choose not to allow Qwiklabs access to your clipboard, you cannot use keyboard shortcuts, but you can still complete the lab.

Code block

Certain steps may include a code block. Click the copy button to copy the code provided and then paste it into the terminal.

sudo apt install nmap

To paste code or other text content that you have copied from the instructions into the terminal, activate the terminal by clicking anywhere inside it. The terminal is active when the cursor in the terminal changes from a static empty outline to a flashing solid block.

gnomecoursewarewikilabs.com wants to see text and images copied to the clipboard. Please click Allow if you would like to be able to use these shortcuts in the Qwiklabs platform. If you choose not to allow Qwiklabs access to your clipboard, you cannot use keyboard shortcuts, but you can still complete the lab.

Once the terminal is active, use the keyboard shortcut **CTRL + V** (hold down the **CTRL** key and press the **V** key) to insert the copied text into the terminal at the location of the flashing cursor.

Scrolling

In certain situations, you may want to scroll within the terminal window. To do so, use the scroll wheel on your mouse or the touchpad of your computer.

End Lab button

End Lab

00:18:05

Finally, click **End Lab** when you've completed the tasks in the lab.

Note: Don't click **End Lab** until you're finished; you'll lose access to the work you've done throughout the lab.

Tracking progress on Coursera

If you complete a lab but your progress hasn't been tracked on Coursera, you may need to refresh the page for your progress to be registered. Once you complete the lab and refresh the page, the green check mark should appear.

Helpful navigation tips and keyboard shortcuts

The following contains a list of navigation tips and keyboard shortcuts you may find useful when completing your Linux labs. Your cursor must be in the terminal window to use these navigation tips and keyboard shortcuts.

- CTRL + C**: Terminates a command that is currently running; from the instructions portion of Qwiklabs, you can use **CTRL + C** to copy, but within the terminal, it will only terminate a command if one isn't running. It will display **^C** at the prompt.
- CTRL + W**: Pastes text.
- clear**: Clears the terminal screen; this can also be done by entering **CTRL + L**.
- CTRL + A**: Sets your cursor at the beginning of a command.
- CTRL + E**: Sets your cursor at the end of a command.
- Left arrow key**: Moves left within a command.
- Right arrow key**: Moves right within a command.
- Up arrow key**: Provides the last command you entered into the command line; can be entered multiple times to go through multiple commands from the command history.
- Down arrow key**: Provides the most command in the command history; must be after using the **Up arrow key**.
- Tab key**: Provides available suggestions for completing your text.

Key takeaways

Knowing how to navigate Qwiklabs will be useful as you complete the labs throughout this course. These labs can help you practice what you've learned in an interactive environment.

Mark as completed

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coach

Hi, Venkatesh!

How can I help?

Explain this topic in simple terms

Give me a summary

Give me practice questions

Give me real-life examples

Ask me anything

6

Go to next item

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https://www.coursera.org/learn/linux-and-sql/supplement/uLa2R/resources-for-completing-linux-labs

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